As-if

```
internal_clock::time_point end=internal_clock::now()+relative_time;
while(!pred())
{
    std::chrono::duration<Rep,Period> remaining_time=
        end-internal_clock::now();
    if(wait_for(lock,remaining_time)==std::cv_status::timeout)
        return pred();
}
return true;
```

Returns

`true` if the most recent call to `pred()` returned `true, false` if the time period specified by `relative_time` has elapsed and `pred()` returned `false`.

Note

The potential for spurious wakeups means that it's unspecified how many times `pred` will be called. `pred` will always be invoked with the mutex referenced by `lock` locked, and the function shall return if (and only if) an evaluation of `(bool)pred()` returns `true` or the time period specified by `relative_time` has elapsed. The thread may be blocked for longer than the specified duration. Where possible, the elapsed time is determined by a steady clock.

Throws

Any exception thrown by a call to `pred`, or `std::system_error` if the effects couldn't be achieved.

Synchronization